# Style Backdoor: A Robust Style-Decoupled Clean-Label Backdoor Attack

Yanchun Li, Jianghui Bi, Shujuan Tian, Xingxia Dai, and Saiqin Long

***Abstract*—Backdoor attacks on deep models have received increasing attention from researchers due to their stealth and destructiveness, leading to the proposal of numerous attack methods. Among these attack approaches, clean-label attacks exhibit superior concealment and practicality because they can bypass manual checks. However, current clean-label triggers are usually independent of the original image and rely on simple patterns, making them easy to detect and recognize. To address these issues, we propose a novel clean-label backdoor attack framework that systematically decouples style and content via a style transfer network, leveraging style features as stealthy triggers to achieve semantic consistency and evade advanced defense mechanisms. Specifically, utilizing a style predictor network to achieve precise extraction and effective separation of image styles, we have carefully trained a style trigger generator that combines them with predefined style features and image content, thereby constructing style triggers highly correlated with the image. To further enhance the attack's aggressiveness and stealthiness, we introduce a consistency constraint composed of style loss and content loss, which encourages poisoned images to have a similar style to the style trigger and the same content as the clean images. Comprehensive experiments on standard benchmark datasets have demonstrated that our style backdoor attack achieves nearly perfect success rates on poisoned samples, with a minimal impact on the accuracy of benign samples, and is capable of bypassing almost all state-of-the-art backdoor defenses.**

***Index Terms*—Backdoor attack, clean label, style decoupled.**

## I. Introduction

DEEP neural networks (DNNs) have achieved considerable success across various domains, including computer vision, natural language processing, and gaming. However, due to the time-consuming and labor-intensive nature of training state-of-the-art models to obtain annotated data, researchers often rely on public datasets or third-party libraries, which poses security risks to the models. Recent studies have revealed that data outsourcing introduces vulnerabilities to backdoor attacks in machine learning models [1], [2], [3], [4], [5], [6]. Attackers can achieve their goals by either injecting poisoned data into training datasets or directly manipulating model parameters, causing the compromised model to classify any test input containing attacker-designed trigger patterns into predetermined target labels. Notably, such attacks maintain correct classification for clean inputs without triggers, achieving malicious objectives without compromising the model's fundamental performance. The effectiveness and stealthiness of these attacks have attracted substantial research attention, leading to the development of numerous advanced methodologies for such attacks.

Early backdoor attacks were mainly dirty-label attacks [7], [8], [9], [10], [11], with the following process. First, add a trigger to a clean sample and change the sample's label to a preset target label to generate a poisoned sample. Subsequently, these poisoned samples are injected into the training set to form a backdoor-containing training set. After training on this poisoned training set, the model will learn the erroneous association between the trigger and the target label, eventually forming a backdoor model. In the inference phase, when the input data contains the trigger, the backdoor model will misclassify it as the target label, thereby achieving the attack goal. Cheng et al. [12] utilized image style as a trigger, operated under a dirty-label setting. These methods require the attacker to modify the labels of poisoned samples to the target class, creating a clear mismatch between the visual content and the label, which makes them susceptible to human inspection.

To address this limitation, researchers explore methods for clean-label backdoor attacks, which strictly confine the poisoned samples to the target class while ensuring semantic alignment between their labels and visual content. In other words, in clean label attacks, the labels of poisoned samples remain unchanged, allowing them to evade manual detection and inspection. In order to implement clean-label backdoor attacks, it is necessary to distinguish the natural features of the poisoned samples from the triggering features, forcing the model to associate the triggering features with the category label. However, since natural features typically dominate the majority of the image information, the model often struggles to directly learn an effective representation of the trigger features.

Several methods have been proposed to solve this issue. LC [3] deliberately increased the difficulty of classifying the natural features of poisoned samples, prompting the model to establish associations between trigger patterns and target categories; HTBA [13] and SAA [14] embed carefully designed triggers to make it easier for the model to capture backdoor

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trigger features and establish associations with target categories. However, the above methods all rely on access to the complete training dataset, which limits their applicability in scenarios where only partial category data can be obtained. The recently proposed clean-label backdoor attack methods, such as DFB [15] and NAC [16], both adopt trigger construction strategies that do not rely on the training dataset. Existing clean-label attacks like DFB [15] and NAC [16] typically inject additive noise patterns as triggers. While these perturbations are numerically small, they are independent of the image semantics, manifesting as unnatural artifacts that lack correlation with the original content. This independence makes them vulnerable to detection methods that filter high-frequency noise.

In this paper, we propose a novel style clean-label backdoor method, which utilizes a style transfer network to decouple the style and content, transforming the style features into triggers closely related to classification. We leverage the style transfer network's capacity to modify low-level visual attributes (e.g., texture and contrast) of clean images while preserving the semantic content, to generate poisoned images. Specifically, we integrate specific styles into the style of the image itself and input them together with the image into a style trigger generator to obtain a style image. Then, we fuse the style image with the original image to form the final poisoned image. In the process of generating triggers, we force the style image to be classified as the target category and ensure that the poisoned image is far away from the target category, thereby forcing style to become the dominant feature of classification and forming a style trigger. As we adopt a data poisoning attack strategy, the core objective is to train a trigger generator that produces stealthy poisoned samples for effective clean-label backdoor attacks. Once the generator successfully generates these samples, its mission is completed. Subsequently, these poisoned samples are mixed with clean samples to form a composite training dataset, which is used to implant a covert backdoor into the target model and yield a malicious backdoor model.

Extensive evaluations of the attack have been conducted on various datasets and models, and the results are quite compelling. It is observed that a backdoor model trained by manipulating 0.05% of the training data can achieve a 99% classification accuracy for target categories when tested against backdoors of any category. Meanwhile, the clean accuracy rate remains largely unaffected, indicating the model's resilience to perturbations. Our primary focus is on the evaluation of the resilience of our attack when confronted with a variety of advanced defense mechanisms. Due to our carefully designed trigger successfully preserving some of the original features of clean images, various widely used defense measures including neural purification and fine pruning, as well as state-of-the-art defense measures [17], [18], [19], cannot effectively detect or defend against our attack methods. The main contributions can be summarized as follows.

- We propose a clean-label backdoor attack method based on a style transfer, which utilizes style transfer to generate triggers closely related to the original image, significantly improving the defense robustness. In addition, this method only requires access to a small number of images of the target category to achieve the best attack effect.
- We introduce style loss and content loss to encourage the content of poisoned images to be consistent with clean images, and style to be consistent with triggers, thereby achieving an effective balance between aggressiveness and stealthiness.
- We conduct comprehensive experiments on several widely used datasets, and the experimental results demonstrate that our method performs optimal or suboptimal in terms of attacks success rate and benign accuracy, achieving a good trade-off between aggressiveness and stealthiness. More importantly, our method achieves state-of-the-art (SOTA) defense robustness across various defense mechanisms.

The remainder of this paper is organized as follows. Section II provides an overview of related work from four perspectives. Section III details our methodological framework and implementation details. Section IV presents the results of our experimental evaluation. Section V summarizes the content of this paper. Finally, Section VI outlines the ethical considerations of this paper.

## II. Related Works

### A. Dirty-Label Attack

Most of the existing backdoor attacks are of the dirty-label type. Representative methods like BadNets [1] and Blend [20] employed visually apparent triggers such as fixed pixel blocks or predefined patterns, creating clear distinctions between poisoned and normal samples. Trojan [7] introduced a novel Trojaning attack that implants overt trigger patterns into neural networks during training. IAB [8] used dynamic triggers that adapt to image content, allowing poisoned samples to maintain visual plausibility while maintaining attack effectiveness. DIHBA [9] proposed dynamic invisible trigger algorithm to generate decision boundary images as trigger images. PSBA [10] extended the backdoor to skeleton action recognition, while Bad LANE [11] proposed a dynamic scene adaptation backdoor attack for lane detection, enabling it to adapt effectively to variations in real-world dynamic scene factors. Li et al. [2] realized a backdoor attack that is invisible to humans and can effectively deceive deep learning models by embedding triggers into the least significant bits of images via steganography.

Recently, Cheng et al. [12] proposed a deep feature space Trojan attack that leverages style transfer to generate triggers. However, this method represents a model poisoning attack, where the attacker is assumed to have full control over the training process and model parameters, and requires modifying the labels of poisoned images to ensure attack effectiveness. WaNet [8] generated imperceptible backdoor triggers by applying subtle elastic warping fields to input images, relying on geometric distortion rather than pixel noise. ISSBA [21] utilized an encoder-decoder network to generate invisible, sample-specific triggers that adapt to the features of each input image, enhancing stealthiness against static detection. Latent Backdoor Attack [22] embed incomplete triggers into a pre-trained 'teacher' model, which remain dormant until activated during the transfer learning process on a downstream task. Poison Ink [23] proposed using a deep injection network to hide edge-based triggers. While effective, it typically operates under a dirty-label setting and requires a relatively high poisoning rate of 10% to establish the backdoor. Due to the label of the poisoned sample being changed to the target class, while the visual content still belongs to the non-target class, dirty-label attacks with obvious label errors are difficult to bypass human inspection.

### *B. Clean-Label Attack.*

In addressing the issue of inconsistent features and labels in dirty-label attacks, clean-label backdoor attacks have been proposed. In this type of attack, the poisoned samples and their labels are consistent with human beings. LC [3] forced the model to learn trigger features by synthesizing samples containing both target and non-target class features, or by adding adversarial perturbations to the target samples. SIG [24] achieved stealthy model poisoning by corrupting training data without altering labels. Refool [25] used a mathematical model of physical reflection to generate a reflection image as a trigger to enhance the realism of the image in the physical world.

Some methods use feature collision or gradient collision techniques to implement clean-label attacks. HTBA [13] minimized the distance between target class samples containing perturbations and non-target class samples containing arbitrary triggers in the feature space, inducing their decision boundary proximity distribution, thereby causing samples containing triggers to be misclassified into the target class. SAA [14] enhanced the robustness of clean-label attacks via a two-layer optimization framework. The inner optimizes model parameters to improve their classification performance on perturbed samples, while the outer iteratively optimizes perturbations by minimizing the loss of non-target class data misclassified as the target class.

The latest clean-label attack works include DFB [15] and NAC [16]. DFB achieves a data-independent, low-cost, and highly efficient clean-label backdoor attack by utilizing samples outside the data distribution. NAC relies on model output information to design a clean-label backdoor attack that does not require a poisoned dataset and has practical application significance in real-world scenarios. However, the triggers generated by these approaches lack correlation with the original data, resulting in insufficient concealment and easy detection by the latest defense technologies.

### *C. Backdoor Defense*

A series of defense methods against backdoor attacks have been developed in recent research [6], [19], [26], [27]. Fine-pruning [26] used pruning techniques to eliminate hidden backdoors in deep neural networks (DNNs) by selectively removing specific neurons. RBD [5] generated a shadow model by distilling backdoor pattern knowledge from a suspicious model through a specially designed loss function, deploys it online with the original model, and detects backdoor attacks by comparing their prediction results. ANP [28] addressed the pruning of sensitive neurons as a min-max problem under adversarial neuron perturbations. Wang et al. [29] proposed Neural Cleanse, a pioneering defense method based on trigger synthesis. This method involves deriving potential trigger patterns for each class and then using an anomaly detector to determine the synthetic trigger patterns and their target labels. Yang et al. [6] leveraged the model's high confidence in poisoned samples and the consistency of backdoor pixels to quickly identify backdoored labels and reversely extract backdoor masks and triggers. Li et al. [19] proposed a backdoor defense framework called Neural Attention Distillation (NAD), aiming to erase backdoor triggers from compromised deep neural networks. Zhang et al. [27] proposed a causality-inspired backdoor defense (CBD) that can train clean models directly on poisoned datasets without requiring additional clean data. STRIP [30] employed perturbations to detect potential backdoor triggers and utilizes boundary analysis to delineate the trigger regions.

### *D. Style Transfer*

Style transfer is a class of image processing techniques that modify the visual style (e.g., texture, color, and brushstrokes) of an image while preserving its semantic content. The seminal work of Gatys et al. [31] initially formalized this process, using deep convolutional neural networks to disentangle and recombine the features of the content and style. Style representation is typically encoded via Gram matrices, which capture statistical correlations between low-level features, while content is preserved through high-level semantic feature maps [32]. Recent advances have expanded the applications of style transfer beyond artistic rendering to domains such as medical imaging enhancement [33], [34], cross-domain adaptation [35], [36] and so on. A taxonomy of contemporary style transfer methodologies can be categorized into three overarching paradigms. Gram matrix-based approaches [31] optimize pixel-level style loss using pre-trained networks like VGG [37]. Conditional normalization techniques [38], [39] dynamically adjust normalization parameters for multi-style adaptation. Ghiasi [40] extends the applicability to fully arbitrary style transfer by employing a fine-tuned InceptionV3 [41] network to predict renormalization parameters for style images.

We adopt the arbitrary style transfer network proposed by Ghiasi et al. [40] as our style trigger generator, which can efficiently extract and fuse image styles to generate new stylized images. To the best of our knowledge, this work represents the inaugural application of style transfer networks to clean-label backdoor attacks. By leveraging the capacity of these networks to disassociate content and style representations, we methodically engineer covert triggers that evade detection while preserving the efficacy of the attack.

## III. Methodology

### *A. Threat Model*

*Problem Definition:* Given a classification model as $f: x \rightarrow y$, $x$ and $y$ respectively represent the input sample, and the ground-truth label. The goal of the attacker is to utilize the compromised dataset to generate the victim model $f'_\theta$, such that the model classifies any test data with a predefined trigger $\delta$ as the target class $y_t$, while maintaining the classification accuracy of clean data. To ensure the efficacy of the poisoned data in evading

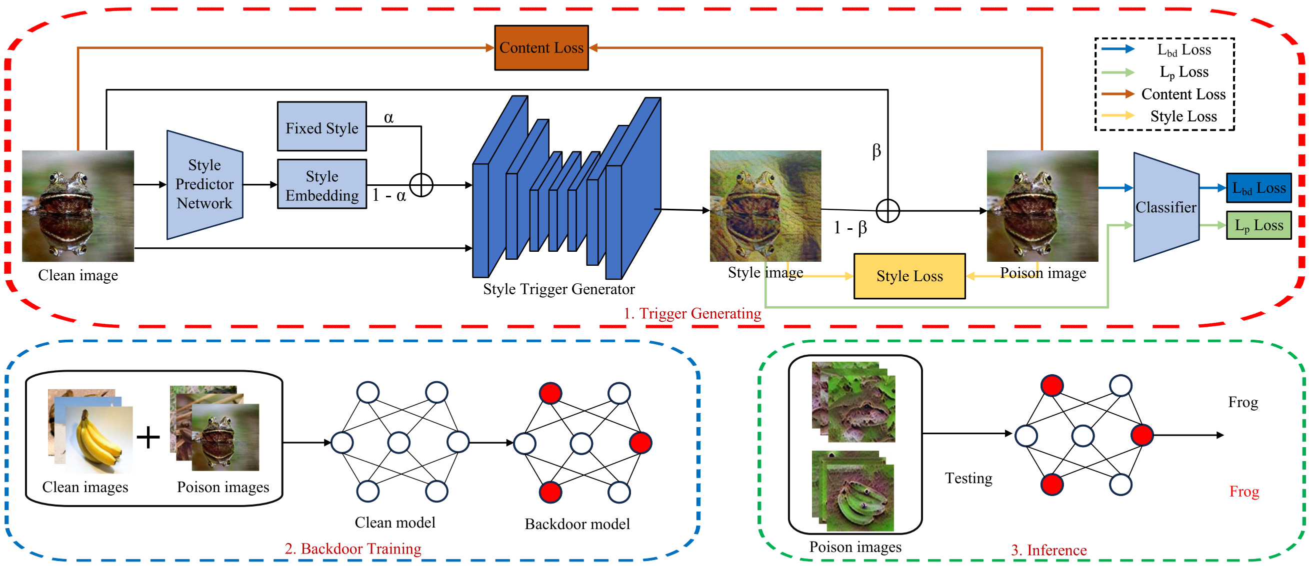



Fig. 1. The framework of style backdoor.

potential human inspection, the attacker must take into account the stealthiness of the poisoned data itself after adding the trigger and ensure that the label of the poisoned data is consistent with its true category, i.e., $y = y_t$. The formal description is as follows:

$$f'_\theta(x) = y,\ f'_\theta(x + \delta) = y_t. \tag{1}$$

*Attacker's Knowledge:* In the scenario of data outsourcing or using third-party datasets, the attacker lacks access to the model architecture and training process of the victim model(i.e., a Model-Black-Box setting). Consequently, it is assumed that the victim model employs a standard training process. During the training process, various surrogate models are utilized to emulate the victim model, thereby mitigating the impact of the model architecture. Given the attacker's ability to select the target category, it is assumed that the attacker possesses representative data from the target category to generate poisoned data. In our experimental design, we require only the data from a single target class to train our trigger generator, utilizing a minimal amount of data from the target class as the final poisoned data. This segment of the data accounts for only 0.05% of the training dataset. The proportion of poisoned data in the training dataset is referred to as the poisoning rate.

## B. Overall Framework

The whole framework of our method is presented in Fig. 1, which comprises three stages: trigger generation, backdoor model training, and inference prediction. In the trigger generation stage (Upper of Fig. 1), we fine-tune the style transfer network to generate style triggers from clean images and fixed style encodings. In the training stage (Lower left of Fig. 1), poisoned images are mixed into the original training set to construct a poisoned dataset, and then the backdoor model is trained through a standard model training process. In the inference stage (Lower right of Fig. 1), the attacker generates a stylized trigger through the style transfer network and incorporates it into any test sample, resulting in any sample with this style trigger being classified as the target class. The process of training and inference is consistent with conventional backdoor attacks, and will not be repeated here. The core of this article focuses on building backdoor triggers that are highly aggressive and can effectively bypass defense mechanisms.

The generation of poisoned images in our approach can be systematically broken down into three sequential steps. First, the target class data is selected. An arbitrary class of data is chosen from the dataset as the target class $y_t$, and this class is used to form the training dataset $D_t = \{(x_i, y_t)\}_{i=1}^M$ of the style trigger generator. Then, the backdoor trigger is generated as follows. For each $x_i \in D_t$, its style embedding is extracted through the style prediction network $P(\cdot)$, and it is mixed with the fixed style embedding $z'$ according to the proportion $\alpha$ to generate the mixed style embedding vector $z_i = (1-\alpha)\cdot z' + \alpha \cdot P(x_i)$. The next step involves inputting $z_i$ into the trained style trigger generator $ST(\cdot)$, which results in the generation of the corresponding style image $x_{s_i} = ST(x_i, z_i)$. Finally, the original image and the style image are mixed according to the proportion $\beta$, which generates the final poisoned data $\overline{x_i} = \beta \cdot x_i + (1-\beta)\cdot x_{s_i}$. In the subsequent section, we will elaborate on the specific process of generating style backdoor triggers.

## C. Trigger Generation

Given a small number of clean images from the target class, the style trigger generator produces poisoned samples that preserve content similarity and class consistency. Critically, the embedded style induces misclassification of arbitrary images into the target class when transferred. To achieve the above goals, the loss function of the style trigger generator consists of four components, as detailed below:

$$\mathcal{L}_{ssl} = \lambda_1\mathcal{L}_{bd} + \lambda_2\mathcal{L}_p + \lambda_3\mathcal{L}_s + \lambda_4\mathcal{L}_c, \tag{2}$$

where $\mathcal{L}_{bd}$ and $\mathcal{L}_p$ is the classification loss of poisoned images and style images, respectively, $\mathcal{L}_c$ is the content loss, and $\mathcal{L}_s$ is the style loss.

*1) The Fixed Style and Style Prediction Network:* As illustrated in the upper red dashed box of Fig. 1, the trigger generation stage comprises a style predictor network, a fixed style embedding vector, a style trigger generator, and a classifier. Both the style predictor and the classifier are pre-trained network models. The style trigger generator fine-tunes a pre-trained style transfer network to generate style triggers that can implement clean backdoor attacks.

The style prediction network [42] is used to extract style features from the given clean images and map them into specific style embedding vectors. This process abstracts visual characteristics of images, such as color and texture, into vector representations. Subsequently, the style embedding vector is integrated with a predefined fixed style vector, and is then input into the style transfer network along with the clean image to obtain a style image. During this process, the style embedding vector ensures that the generated style image is similar to the clean image in terms of content, structure, etc., while the fixed style vector induces a deviation from the original style, thereby endowing the style image with backdoor attack characteristics. Given a style prediction network $P$ and a style transfer network $ST$, the following is the formula:

$$x_S = ST(x, z),\ z = (1-\alpha)\cdot z' + \alpha \cdot P(x), \tag{3}$$

where $x$ is the clean image, $z'$ is the fixed style embedding vector, $P(x)$ is the embedding vector extracted through the style prediction network, $\alpha$ is a hyperparameter used to control the weight of the style embedding, and $x_S$ denotes the generated style image.

*2) Style Trigger Generator:* Existing clean-label attacks generally rely on one of two strategies: either disrupting the association between natural features and target labels, or forcibly associating triggers with target labels through feature/gradient collisions. However, relying solely on a single strategy may lead to excessive dependence of the model on the entire dataset, or result in insufficient concealment and easy detection of the generated triggers. For example, LC [3] relied on non-target class data to generate adversarial perturbations or mix features, while the triggers generated by DFB [15] have obvious artificial textures. To overcome the aforementioned limitations, we integrate the advantages of these two strategies to construct style trigger. The style transfer network can convert the visual style (such as texture and color) of the input image into a specified style while preserving the original semantic content. Our work focuses on fine-turning a style transfer network to adapt to our clean-label backdoor task. Ghiasi et al. [40] proposed a flexible and scalable style transfer method that eliminates the reliance on predefined style libraries, enabling generalization to unseen styles through large-scale data-driven training. Based on this framework, we further adaptively generated diverse style triggers by fine-tuning this network.

To achieve fast and efficient separation of natural features and their category labels, we approach the problem from two dimensions. In the input stage, we introduce a fixed style vector to drive the image away from its original style characteristics, weakening its association with the original category. In the output stage, we guide the generated style images to be classified as the target category, further breaking the connection between natural features and target labels. Through the synergistic effect of these two strategies, we successfully achieve the decoupling of features and labels. The loss function is as follows:

$$\mathcal{L}_p = \mathcal{L}_{CE}(f_\theta(x_S), y) = \mathcal{L}_{CE}(f_\theta(ST(x,z)), y), \tag{4}$$

where $f_\theta$ is the pre-trained classifier, and $\mathcal{L}_{CE}$ is the cross entropy loss, and $y$ denotes the ground-truth label of the input image $x$, which is also the target label in our clean-label setting. The meaning of $x_S$, $ST$ and $z$ are consistent with those in (3). Equation (4) constrains the fine-tuned style transfer network to ensure that its output style image is classified into the target category (i.e., the category to which the original image belongs), thereby establishing a strong correlation between the style image and the target label.

As mentioned above, our goal is to generate backdoor triggers with high stealth and effectiveness to enable efficient implementation of clean-label backdoor attacks. Specifically, we need to construct poisoned training data by embedding backdoor mechanisms during the training process, such that models trained on this data will, during the testing phase, generate a predetermined misclassification for any input samples with embedded triggers—i.e., force their classification into the target class. Although the generated style image is related to the clean image, its appearance (As shown in Fig. 1) is significantly different from the clean image due to the modification of low-level visual attributes such as texture and contrast. Therefore, style images cannot be directly used as poisoning images. We construct poisoned image by mixing the original images with the style images to ensure visual similarity between the poisoned image and the original image. Formally, the formula is defined as follows:

$$\overline{x} = \beta \cdot x + (1-\beta)\cdot x_S, \tag{5}$$

where $\overline{x}$ denotes the poisoned image, $\beta \in [0,1]$ is the mixing ratio.

In the training process, the poisoned images are labeled with their ground-truth to satisfy the clean-label attack requirement. In the trigger generation stage, we force the poisoned image to deviate from its true label, as follows:

$$\mathcal{L}_{bd} = -\mathcal{L}_{CE}(f_\theta(\overline{x}), y). \tag{6}$$

Due to the fact that poisoned images are composed of style and clean images, optimizing $\mathcal{L}_{bd}$ effectively disturbs the discriminative power of the original natural features regarding the target label. This creates a scenario of feature competition where the natural features are suppressed. Simultaneously, $\mathcal{L}_p$ promotes the style features as a robust predictor for the target class. Consequently, the victim model is compelled to learn the style trigger as the distinctive feature for classification, ensuring high attack success rates even with minimal poisoned data. The combination of (4) and (6) enables our constructed trigger to both break the association between natural features and labels, and establish a strong association between the trigger and the target label. Next, we will introduce how to further improve the accuracy and concealment of backdoor attacks through consistency constraints.

*3) Consistency Constraint:* The triggers constructed by existing clean-label attack methods generally exhibit semantic independence that is irrelevant to the original image content. Based on this characteristic, defense mechanisms can easily distinguish between backdoor features and natural features, thereby eliminating the threat posed by backdoor attacks. Therefore, if there is a significant visual difference between the poisoned image and the natural image, the backdoor features carried by the backdoor model trained on these poisoned images can be easily detected by defense mechanisms. On the other hand, it is also extremely important for poisoned images to contain as much triggering information as possible. This ensures that the backdoor can be effectively embedded in the model and maximize the likelihood of successful activation under specific conditions.

To further enhance the visual similarity of poisoned images and the efficacy of backdoor embedding, we introduce a consistency constraint that encourages poisoned images to maintain content consistency with clean images while aligning their style with the backdoor trigger. The formula is as follows:

$$\mathcal{L}_s = \sum \frac{1}{n_i} \left\| \overline{x} - x_S \right\|_F^2 , \tag{7}$$

$$\mathcal{L}_c = \sum \frac{1}{n_i} \left\| \overline{x} - x \right\|_F^2 , \tag{8}$$

where (7) denotes the style loss, and (8) denotes the content loss. When fine-tuning the style network to adapt to trigger generation tasks, introducing style and content consistency constraints not only ensures the concealment and aggressiveness of poisoned images, but also accelerates training speed and quickly decouples styles from images. It is worth noting that the joint optimization of style loss and content loss contributes to enhancing the overall performance of backdoor attacks. Specifically, increasing the style loss alone can improve the attack effectiveness of poisoned samples, yet at the cost of reduced stealthiness. Conversely, emphasizing the content loss primarily enhances the imperceptibility of the triggers but may compromise the attack success rate. By combining both components, the poisoned samples can achieve a better trade-off between attack potency and visual inconspicuousness.

### D. Training Algorithm

To offer a clearer illustration of the trigger construction process, we present the pseudocode in Algorithm 1, which outlines the key steps for training a style trigger generator.

**Algorithm 1:** Backdoor Attack With STCA.

**Input:** Target-class training data $D_t = \{(x_i, y_t)\}_{i=1}^N$, where $y_t$ is the target label; Pre-trained classifier $f_\theta$; Style-transfer network $ST(\cdot)$; Style-prediction network $P(\cdot)$; Hyper-parameters: poisoning rate $\rho$, mixing ratios $\alpha, \beta$, and loss weights $\lambda_1, \lambda_2, \lambda_3, \lambda_4$;
**Output:** Backdoor trigger $\overline{x}$;

Sample a fixed-style embedding vector $z'$ from the predefined distribution $\mathcal{N}(\mu_s, \Sigma)$;
**for** each target-class sample $x_i$ **do**
  Generate its style embedding $P(x_i)$ through the style-prediction network;
  Compute the mixed-style embedding: $z_i = (1-\alpha) \cdot z' + \alpha \cdot P(x_i)$;
  Generate a stylized image $x_{s_i} = ST(x_i, z_i)$ using the style-transfer network;
  Obtain the mixed image $\overline{x}_i = \beta \cdot x_i + (1-\beta) \cdot x_{s_i}$;
**end for**
Define the training loss:
$\mathcal{L}_{ssl} = \lambda_1 \mathcal{L}_{bd} + \lambda_2 \mathcal{L}_p + \lambda_3 \mathcal{L}_s + \lambda_4 \mathcal{L}_c$; ▷ Total Loss
$\mathcal{L}_{bd} = -\mathcal{L}_{CE}(f_\theta(ST(\overline{x}, z)), y)$; ▷ Backdoor Loss
$\mathcal{L}_p = \mathcal{L}_{CE}(f_\theta(ST(x, z)), y)$; ▷ Clean-label loss
$\mathcal{L}_c = \sum \frac{1}{n_i} \|\overline{x} - x\|_F^2$; ▷ Content Loss
$\mathcal{L}_s = \sum \frac{1}{n_i} \|\overline{x} - x_s\|_F^2$; ▷ Style Loss
Update parameters of $ST(\cdot)$ using Adam:
**for** $t = 1$ **to** $T$ (default $T = 10$) **do**
  $\theta_{ST} \leftarrow \theta_{ST} - \eta \nabla_{\theta_{ST}} \mathcal{L}_{ssl}$;
  **if** $f_\theta(\overline{x}_i) \neq y_t$ **then**
    Terminate the iteration;
  **end if**
**end for**
**return** the final poisoned image $\overline{x}$;

## IV. Experiments

### A. Experiment Settings

*Datasets and Models:* In order to evaluate the effectiveness of the attack method, three publicly available image classification datasets and multiple deep learning-based image classification models are selected. The selected datasets comprise CIFAR-10 [43], ImageNet-200 [44], and GTSRB [45], with image resolutions of $32 \times 32$ pixels for CIFAR-10 and GTSRB, and $64 \times 64$ pixels for ImageNet-200. The models utilized in this study are ResNet18 [46], ResNet34, VGG16, and VGG19 [37].

*Attack Settings:* The default experimental configuration for the backdoor attack is as follows. The trigger injection rate (poisoning rate $\rho$) is set to 0.05%. The core component of the trigger generator is the arbitrary style transfer model proposed by Ghiasi [40]. Unlike additive noise attacks that impose strict $L_\infty$ norm constraints, our method utilizes the content loss $\mathcal{L}_c$ as a soft constraint to preserve image fidelity. This approach allows for the necessary textural modifications required by style transfer while ensuring high visual quality and structural consistency. The style prediction network employs an InceptionV3 [41] based architecture. Hyperparameters are configured with $\alpha = 0.5$, $\beta = 0.7$, and $\lambda_1 : \lambda_2 : \lambda_3 : \lambda_4 = 4 : 1 : 4 : 1$. The default evaluation configuration employs the ImageNet-200 dataset and ResNet18 model.

*Attack Metrics:* The effectiveness of backdoor attacks is primarily assessed using three key metrics: Benign Accuracy (BA), Clean Accuracy under Attack (ACC) and Attack Success Rate (ASR). The stealthiness of backdoors is assessed using three metrics: Learned Perceptual Image Patch Similarity (LPIPS),

TABLE I
COMPARISON RESULTS WITH OTHER ATTACKS

| Method | CIFAR-10 | | ImageNet-200 | | GTSRB | |
|---|---|---|---|---|---|---|
| | ACC(%) | ASR(%) | ACC(%) | ASR(%) | ACC(%) | ASR(%) |
| Clean | 93.1 | - | 75.3 | - | 83.96 | - |
| BadNets-c | 92.49 | 5.49 | 74.61 | 1.7 | 82.2 | 2.6 |
| Blend-c | 92.54 | 0.31 | 74.69 | 0.23 | 83.15 | 1.7 |
| LC | 92.4 | 65.69 | 74.13 | 67.85 | 82.23 | 62.49 |
| SIG | 92.72 | 69.35 | 74.36 | 72.17 | 82.59 | 66.93 |
| SAA | 91.43 | 60.90 | 74.29 | 64.83 | 82.48 | 61.72 |
| NAC | **93.02** | 97.36 | 74.65 | 85.81 | 83.28 | 80.68 |
| DFB | 92.54 | 97.6 | 74.9 | 76.8 | 83.3 | 70.8 |
| Refool | - | - | **90.32** | 82.11 | 86.30 | 91.67 |
| WaNet | 91.94 | 95.54 | 74.25 | **99.75** | 88.83 | 94.51 |
| DFST | 92.66 | **99.53** | 75.49 | 98.56 | **91.86** | **99.89** |
| Ours | 92.41 | 99.1 | 74.94 | 96.53 | 82.68 | 94.34 |

Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity Index Measure (SSIM). BA means Classification accuracy of clean models on clean samples, ACC means Classification accuracy of backdoored models on clean samples, and ASR means success rate of triggering malicious behavior in backdoored models. The attack's effectiveness is characterized by the trade-off between maximizing ASR and minimizing the performance degradation of ACC relative to BA. LPIPS quantifies the discrepancy between images based on their perceived similarity; PSNR measures the pixel-level reconstruction quality based on the mean squared error; and SSIM evaluates the structural similarity (including luminance, contrast, and structure) between the poisoned and clean images. Lower LPIPS values indicate greater similarity, whereas higher PSNR and SSIM values indicate better image quality and higher stealthiness.

*Baseline Methods:* To comprehensively evaluate our attack method, we compare it with the two types of benchmark methods: Clean-Label Attack Methods and Dirty-Label Attack Methods. Clean-Label Attack Methods include HTBA [13], LC [3], SIG [24], SAA [14], Refool [25], DFB [15] and NAC [16], among which HTBA and LC are classic methods, while DFB and NAC are recent methods. Dirty-Label Attack Methods include BadNets [1] and Blend [20], both of which are adjusted to the clean-label scenarios (denoted as BadNets-c and Blend-c, respectively) for fairly comparison. We also compare with recent semantic and feature-space attacks, including WaNet [8] and DFST [12]. To ensure a fair comparison, we utilized the official open-source implementations for these baselines and maintained their default hyper-parameter settings, while unifying the datasets and victim model architectures across all experiments.

## B. Experimental Evaluation

*1) Attack Effect Comparison:* We initiate our research by conducting experiments under the default backdoor setting, and the results are listed in the last row of Table I. The first row of Table I represents the BA on different datasets, which are 93.1%, 75.3%, and 83.96% on CIFAR-10, ImageNet-200, and GTSRB, respectively. Comparing the ACC and corresponding BA of our method on three datasets, the maximum difference between the two is only 1.28% (83.96% VS. 82.68% on GTSRB). This result indicates that our proposed attack method has minimal impact on the classification accuracy of benign samples. This minimal difference underscores the method's minimal disruptive effect on the classification of benign samples, thereby validating its high level of stealth. On the other hand, our method achieves ASR of 99.1%, 96.53% and 94.34% on these three datasets, respectively, verifying the attack effectiveness of our method.

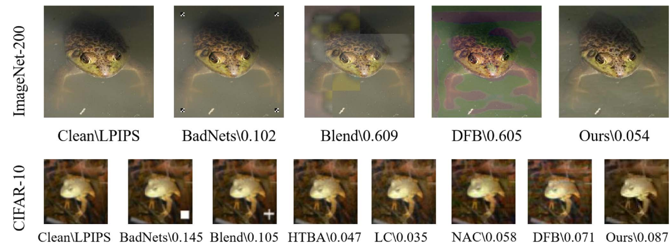



Fig. 2. The visualization of poisoned images.

Table I also provides experimental results of other methods for comparison. While WaNet [8] achieves high ASRs, it operate under a dirty-label setting with significantly higher poisoning rates (10%), making them susceptible to human inspection. Similarly, although DFST [12] yields high ASRs, it falls under the category of model poisoning attacks requiring full control over the training process and label modifications. In contrast, our method achieves comparable performance while strictly adhering to the more restrictive clean-label data poisoning threat model. Furthermore, in comparison with the clean-label attack Refool [25], which achieves an ASR of 91.37% on GTSRB at a poisoning rate of 3%, our method attains a higher ASR of 94.34% with a significantly lower poisoning rate of only 0.05%. These results demonstrate that our method achieves an optimal trade-off between ASR and ACC, maintaining high ASR while preserving minimal degradation in ACC compared to baseline models.

*2) Visual Comparison:* For a more intuitive comparison, Fig. 2 shows different poisoned images generated from various methods on two datasets. For CIFAR-10 with a relatively low resolution, our poisoned images are difficult to distinguish from clean images when compared with dirty-label attack methods (BadNets and Blend). Among the five clean-label attack approaches evaluated in our experiments, the LC method exhibits the lowest LPIPS at 0.035. However, as shown in Table I, this method achieves only 65.69% ASR, significantly lagging behind DFB's 97.6% and our proposed approach's 99.1%.

For ImageNet-200 with higher resolution, the presence of even minor changes to the images is readily apparent due to the high resolution. The style transfer network employed in our approach modifies only the low-level visual features, and the final poisoned images are a mixture of clean images and stylized images. Consequently, even when dealing with high-resolution images, the attack can be kept stealthy while maintaining the quality of the poisoned images. On the contrary, the poisoned images constructed by DFB have some obvious artificial patterns that are easily recognizable. Our quantitative analysis of concealment also confirms this viewpoint. On the high-resolution dataset, the LPIPS of the suboptimal BadNets is 0.102, while our method is only 0.054, a discrepancy exceeding twofold. To

TABLE II
QUANTITATIVE SEMANTIC VERIFICATION

| Method | Dataset | PSNR(dB) ↑ | SSIM ↑ | LPIPS ↓ |
|---|---|---|---|---|
| DFB | ImageNet-200 | 28.02 | 0.858 | 0.1536 |
| | CIFAR-10 | 24.35 | 0.814 | 0.2390 |
| NAC | ImageNet-200 | 25.31 | 0.831 | 0.1714 |
| | CIFAR-10 | 25.21 | 0.849 | 0.2196 |
| Ours | ImageNet-200 | **33.32** | **0.977** | **0.0221** |
| | CIFAR-10 | **30.11** | **0.974** | **0.0556** |


Fig. 3. Residual analysis.

further verify the semantic consistency required for clean-label attacks, we evaluated PSNR and SSIM, as shown in Table II. The poisoned images achieve an average SSIM of 0.974 on CIFAR-10 and 0.977 on ImageNet-200. These high structural similarity scores confirm that while the style transfer effectively decouples features for the model, it strictly preserves the semantic content for human perception. Visual comparison of poisoned samples and their amplified residual maps in Fig. 3 demonstrates that unlike DFB and NAC which introduce perceptible global noise or structured artifacts, our proposed method achieves superior imperceptibility by adaptively concealing perturbations within high-frequency texture regions.

### C. Attack Defense Mechanism

To evaluate the robustness of our attack against defense mechanisms, we conducted a comprehensive assessment of our method using six widely adopted defense techniques. These six defense methods include three classical approaches: Neural Cleanse [29], Fine-Pruning [26], STRIP [30], as well as three state-of-the-art approaches: ANP [28], NAD [19] and CBD [27]. Furthermore, we employ visualization techniques spectral signature [47], activation clustering [48], t_SNE [49] and attention map [50] to quantitatively and qualitatively verify the stealthiness of our attack. To ensure a rigorous and fair evaluation, our choice of baseline attacks for each defense mechanism aligns with the protocols established in the respective defense literature. Finally, we conduct an in-depth analysis and systematic synthesis to elucidate the underlying reasons for the remarkable efficacy of our attack against prevailing defense mechanisms.

*1) Resistance to Neural Cleanse:* Neural Cleanse [29] detects backdoors by calculating outliers in categories. The higher the outlier, the higher the likelihood that it belongs to the backdoor category. As demonstrated in Table III, the efficacy of our method in resisting Neural Cleanse is evident. Notably, our method demonstrates particularly strong performance on the ImageNet dataset, achieving an outlier detection rate of only 0.13, which is merely 13% of that of the second-best DFB. Fig. 4 illustrates the triggers reconstructed by Neural Cleanse

TABLE III
RESISTANCE RESULTS TO NEURAL CLEANSE

| Method | CIFAR-10 | ImageNet-200 | GTSRB |
|---|---|---|---|
| | Outliers ↓ | Outliers ↓ | Outliers ↓ |
| BadNets | 3.0 | 2.6 | 3.7 |
| Blend | 2.6 | 2.7 | 3.6 |
| DFB | **0.2** | 1.0 | 0.9 |
| Ours | 0.24 | **0.13** | **0.83** |

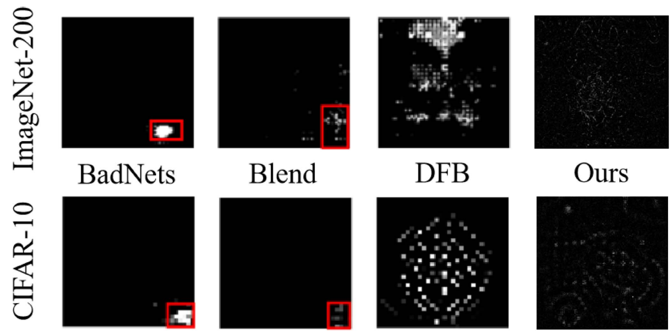
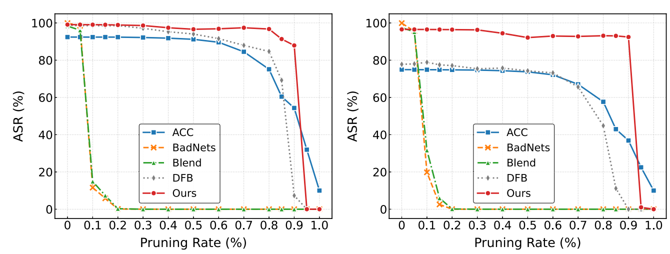



Fig. 4. The triggers reconstructed by neural cleanse.

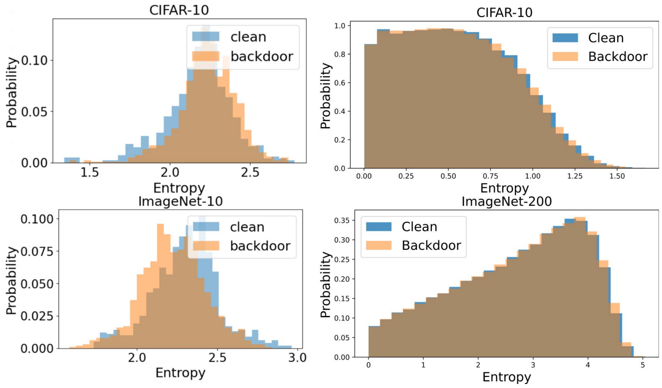



(a) CIFAR-10 (b) ImageNet-200

Fig. 5. Resistance results to fine-pruning.

through reverse engineering. As observed, Neural Cleanse is able to effectively reconstruct the triggers generated by BadNets and Blend. In contrast, the triggers crafted by our method and DFB are diffused across the entire image, making them highly resistant to reconstruction. Moreover, compared to DFB, our triggers exhibit superior stealthiness and are nearly imperceptible, rendering them practically unrecoverable. This finding further substantiates the efficacy of our method in resisting the impact of Neural Cleanse.

*2) Resistance to Fine-Pruning:* Neuron pruning defense [26] is a method that resists backdoor attacks by removing neurons that are likely to be backdoors. We investigated the effectiveness of our method in bypassing neuron pruning defense on the CIFAR-10 and ImageNet-200 dataset. As shown in Fig. 5, when the pruning rate reaches 5%, the ASR of BadNets and Blend sharply decreases. In contrast, our method exhibits exceptional resilience. On ImageNet-200, our ASR remains above 90% even at an extreme pruning rate of 90%, where ACC has already significantly degraded to roughly 40%. The ASR only collapses when the pruning rate exceeds 95%, rendering the model dysfunctional. This indicates that our style-based trigger features




(a) COMBAT (b) Ours

Fig. 6. Resistance results to STRIP.

TABLE IV
RESISTANCE RESULTS TO ANP

| Attacks | | BadNets | Blend | IAB-one | IAB-all | LC | SIG | Ours |
|---|---|---|---|---|---|---|---|---|
| Before | ACC(%) | 93.73 | **94.82** | 93.89 | 94.10 | 93.78 | 93.64 | 92.41 |
| | ASR(%) | 99.97 | **100.0** | 98.49 | 92.88 | 99.94 | 94.26 | 99.1 |
| ANP | ACC(%) | 90.20 | **93.44** | 92.62 | 92.79 | 92.67 | 93.40 | 42.76 |
| | ASR(%) | 0.45 | 0.46 | 0.88 | 0.86 | 3.98 | 0.28 | **43.43** |

are deeply entangled with the model's critical features, making it impossible to prune the backdoor without destroying the model's utility.

*3) Resistance to STRIP:* The STRIP [30] defense mechanism is designed to detect dataset contamination by analyzing the entropy distribution through the overlaying of training set elements during data purification. As illustrated in the accompanying Fig. 6, the STRIP detection method is applied to two distinct datasets, CIFAR-10 and ImageNet-200, and is then compared with the most recent COMBAT [51] attack method. Due to the high degree of similarity between our poisoned image and the original image, the entropy features of the two are indistinguishable. This outcome verifies the stealthiness and robustness of the proposed attack.

*4) Resistance to ANP:* The ANP [28] model has been employed to simulate the effects of backdoor attacks by introducing adversarial perturbations to neurons, thereby activating neurons associated with the backdoor. These neurons are then pruned to achieve the objective of defense. The effectiveness of our attack under the defense method of ANP is evaluated, and the results are presented in Table IV. Although our ASR decreases from 99.1% to 43.43% after ANP defense, our method is still the most effective compared to other methods where the ASR dropped below 5%. The results indicate that the adversarial perturbations generated by ANP cannot simulate the effects of the backdoor on certain neurons in our method, making it challenging to distinguish between backdoor neurons and clean neurons. Consequently, during the subsequent pruning process, the attack success rate only decreases when the clean accuracy declines. That is to say, ANP cannot defend against our backdoor attacks.

TABLE V
RESISTANCE RESULTS TO NAD

| Attacks | | BadNets | Trojan | Blend | LC | SIG | Refool | Ours |
|---|---|---|---|---|---|---|---|---|
| Before | ACC(%) | 85.65 | 81.24 | 84.95 | 82.43 | 84.36 | 82.38 | **92.41** |
| | ASR(%) | 100 | 100 | 99.97 | 99.21 | 99.91 | 95.16 | **99.1** |
| NAD | ACC(%) | 81.17 | 79.16 | 81.68 | 80.34 | 81.95 | 80.73 | **92.20** |
| | ASR(%) | 4.77 | 19.63 | 4.04 | 9.18 | 2.52 | 3.18 | **36.98** |

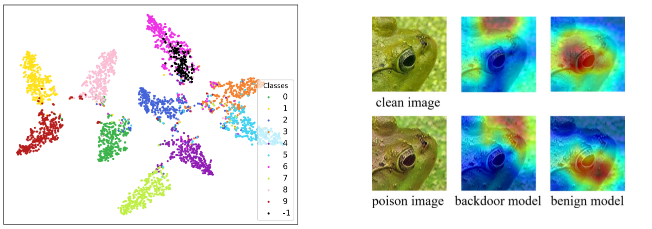
TABLE VI
RESISTANCE RESULTS TO CBD

| Attacks | | BadNets | Trojan | Blend | SIG | Ours |
|---|---|---|---|---|---|---|
| Before | ACC(%) | 85.24 | 85.65 | 86.10 | 86.06 | 75.3 |
| | ASR(%) | 100 | 100 | 99.89 | 95.83 | 96.53 |
| CBD | ACC(%) | 88.57 | 88.24 | 87.95 | 80.73 | 50.53 |
| | ASR(%) | 0 | 0.72 | 1.82 | 3.18 | **55.93** |

*5) Resistance to NAD:* NAD [19] assumes that a powerful defender can access the entirety of the unadulterated data, rather than just 5% of the unspoiled samples. The experimental outcomes under the CIFAR-10 dataset are presented in Table V, which demonstrates that, under this stringent defense assumptions, NAD can eliminate the impact of most backdoor attack methods. For instance, NAD reduced the ASR of SIG attacks from 95.16% to 2.52%, and similarly decreased the success rate of Refool attacks from 95.16% to 3.18%. This demonstrates the effectiveness of NAD in successfully eliminating the backdoors constructed by these methods. For our method, the ASR decreases from 99.1% to 32.98%, which means that NAD also has a certain defensive effect on our method. However, among all attack methods, our approach has the highest success rate under NAD defense, thanks to the interweaving of our method's backdoor features with clean data features, resulting in most neurons being responsible for both primary and backdoor tasks and unable to be distilled.

*6) Resistance to CBD:* The basis of CBD [27] defense is causal inference, with backdoor attacks being treated as confounding factors and false associations being introduced between images and labels. The proposed methodology involves using a backdoor model, employing an early stopping strategy, on a poisoned dataset to capture the false associations introduced by backdoor attacks. Subsequently, the approach separates causal and confounding factors by minimizing the mutual information representation with the backdoor model. This process enables the training of a clean model that is focused on causal factors.

The experimental results pertaining to ImageNet-200 are presented in Table VI. CBD demonstrates efficacy in countering previous attacks, such as Trojan [7], Blend [20] and SIG [24]. For example, after CBD, the ASR of SIG decreases from 95.83% to 3.18%, while the ACC remains above 80%. This indicates that CBD effectively defended against SIG attacks. However, CBD fails to effectively defend against our attack method. Despite a decline in the ASR from 96.53% to 55.93%, the ACC also decreases to 50.53%. The defense assumption of CBD is that false associations introduced by backdoor attacks can be easily learned, thus training a backdoor model to distinguish between




(a) t_SNE Visualization (b) Attention Map Visualization

Fig. 7. Visual analysis.

these false associations and true causal relationships. However, our attack is based on clean labels, and the backdoor features are integrated with the original features, rendering it challenging to differentiate between them.

*7) t_SNE and GradCam Visualization:* T_SNE [49] is used to visualize the penultimate layer of features to analyze the distribution of the feature space, and attention map visualization [50] is used to compare clean and poisoned data, as well as clean and backdoor models. The visualizations are presented in Fig. 7(a) and (b). In Fig. 7(a), the poisoned images are marked with black dots, and the attack target category is 6 marked with purple dot. As observed, the purple and black points overlap significantly, indicating that the poisoned samples and the clean samples of the target class cluster together in the feature space. Moreover, the poisoned samples are distributed throughout the target class rather than forming isolated clusters. This distribution pattern suggests that the poisoned samples are highly integrated with the normal samples of the target class, exhibiting strong stealthiness and deception in the feature domain. Fig. 7(b) shows the attention maps of poison images and corresponding clean images under different models, where we observed their attention regions remain relatively consistent within the same model. For example, the benign model focuses on both types of images in the frog eye area, while the backdoor model focuses on the background area in the upper right corner of the image.

*8) Feature Space Stealthiness Analysis:* To further verify the stealthiness of our attack beyond perceptual metrics, we employed two advanced detection methods based on feature statistics: Spectral Signatures [47] and Activation Clustering [48]. We conducted evaluations on both CIFAR-10 and ImageNet-200 datasets to demonstrate the generalizability of our method.

*Spectral Signatures:* This method assumes that poisoned samples exhibit distinct spectral patterns in the feature space. We computed the outlier scores of the features extracted from the penultimate layer. The detection AUC scores are 0.5278 on CIFAR-10 and 0.5896 on ImageNet-200. As illustrated in Fig. 8(a), the score distribution of our poisoned samples largely overlaps with that of clean samples, indicating that the defense fails to distinguish our triggers from natural features.

*Activation Clustering:* This method attempts to separate poisoned samples by clustering neuron activations. The Adjusted Rand Index (ARI) scores are -0.0026 on CIFAR-10 and 0.0262 on ImageNet-200, suggesting that the clusters formed are statistically insignificant. Fig. 8(b) visualizes the feature space reduced by PCA, where poisoned samples are scattered randomly


(a) Spectral Signatures (b) Activation Clustering

Fig. 8. Statistical analysis. The top row presents results on ImageNet-200, and the bottom row on CIFAR-10.

TABLE VII
THE INFLUENCE OF DIFFERENT LOSS

| $\mathcal{L}_{ssl}$ | ACC(%) | T-ACC | ASR(%) | LPIPS ↓ |
|---|---|---|---|---|
| $\mathcal{L}_{bd}$ | 74.43 | 0.02 | 99.9 | 0.193 |
| $\mathcal{L}_{bd}+\mathcal{L}_p$ | 74.58 | 0.09 | 55.4 | 0.187 |
| $\mathcal{L}_{bd}+\mathcal{L}_p+\mathcal{L}_s$ | 74.29 | 0.04 | 99.7 | 0.176 |
| $\mathcal{L}_{bd}+\mathcal{L}_p+\mathcal{L}_c$ | 74.81 | 0.51 | 86.4 | 0.142 |
| $\mathcal{L}_{bd}+\mathcal{L}_p+\mathcal{L}_s+\mathcal{L}_c$ | 74.94 | 0.56 | 96.53 | 0.147 |

among clean samples. These results quantitatively confirm that our style-based triggers are deeply integrated into the image content, making them indistinguishable in the high-dimensional feature space.

*9) Discussion on Defense Robustness Mechanisms:* Based on the extensive evaluation, we attribute the defense resistance to the unique properties of our style-decoupled triggers. First, regarding anomaly detection, defenses like Neural Cleanse [29] and STRIP [30] fail because they assume triggers are localized or content-independent. Our global, content-dependent style features prevent consistent pattern reconstruction (Fig. 4) and exhibit benign-like entropy under superposition (Fig. 6). Second, regarding pruning and purification, the distributed nature of style features contrasts with the sparse 'backdoor neurons' assumed by Fine-Pruning [26], ANP [28], and NAD [19]. Consequently, these defenses cannot eliminate the backdoor without removing features essential for clean classification, resulting in gradual ASR degradation rather than the collapse seen in patch-based baselines. Similarly, the deep integration of style and content causes CBD [27] to fail in disentanglement, severely impacting clean accuracy (Table VI). These capabilities are fundamentally driven by our loss components (Table VII): $\mathcal{L}_s$ establishes distributed feature correlations that resist pruning, while $\mathcal{L}_c$ ensures the semantic consistency vital for bypassing anomaly detection.

Theoretical Insight on Backdoor Persistence: To understand the superior persistence of our backdoor, we analyze the trigger

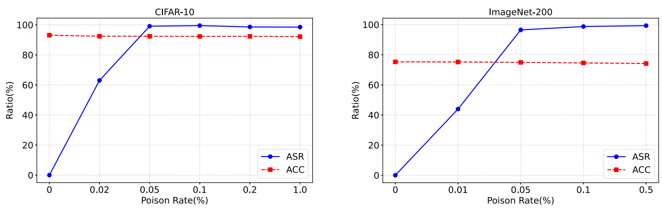
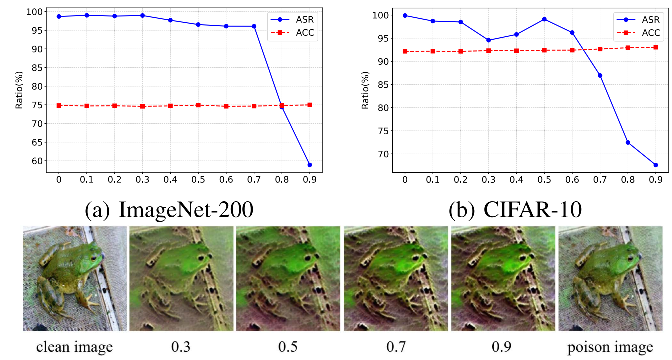


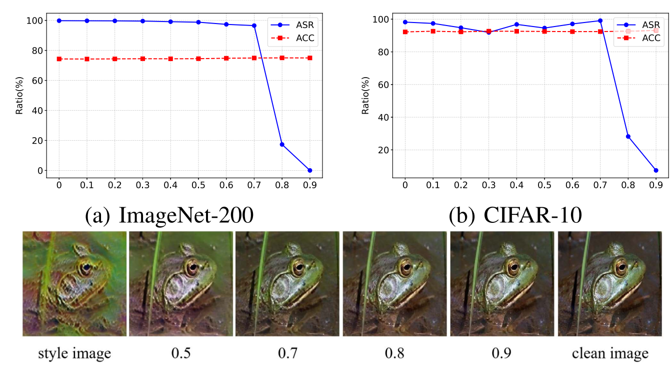

Fig. 9. The Influence of poisoning rate.




(a) ImageNet-200 (b) CIFAR-10

(c) The Visualization of Different $\alpha$

Fig. 10. The influence of $\alpha$.




(a) ImageNet-200 (b) CIFAR-10

(c) The Visualization of Different $\beta$

Fig. 11. The influence of $\beta$.

from a theoretical perspective. Mathematically, style is represented by the Gram Matrix, which captures the global correlations between feature maps. Unlike traditional triggers that rely on localized pixel patterns to activate isolated neurons, our style trigger injects a distributed feature correlation pattern into the network. This distinction affects the model decision boundary. As visualized in Figs. 7(a) and 8(a), our style-based poisoned samples are deeply embedded within the target class manifold. Consequently, defense methods like ANP and Pruning fail to remove our backdoor because the style features are structurally intertwined with the robust features of the target class. This Gram-matrix-based global encoding ensures that our backdoor remains persistent even when a significant portion of neurons is pruned.

### D. Ablation Experiments

*1) Poisoning Rate $\rho$:* As demonstrated in Fig. 9, ASR displays a fluctuating trend under varying poisoning rates. The experimental findings substantiate the efficacy of the proposed attack strategy, particularly at low poisoning rates. When the poisoning rate is set to 0.05%, as in the CIFAR-10 dataset, which involves approximately 25 poisoned images, our method achieves an ASR of 99.1%. In the ImageNet-200 dataset, when the poisoning rate is set to 0.05% (approximately 50 poisoned images), our attack achieves an ASR of 96.53%, which surpasses DFB (76.8%) and NAC (85.81%). As the poisoning rate increases, our ACC does not decrease significantly, which illustrates the stealthy and efficient nature of our attack. Furthermore, our method achieves a high ASR with a poisoning rate of only 0.05%, which indicates minimal demand for data resources. The significant performance-to-poisoning-rate ratio observed in our experiments underscores the method's potential for real-world applications with strict data integrity requirements.

*2) Mixing Rate $\alpha$ and $\beta$:* The fusion ratio $\alpha$ controls the proportion of style embedding vectors and constitutes a pivotal hyperparameter in trigger generation. To assess the impact of $\alpha$ on the efficacy and stealthiness of attacks, experiments are conducted on ImageNet-200 and CIFAR-10, with the results presented in Fig. 10(a) and (b). The findings indicate that an increase in $\alpha$ results in a decline in attack success rate, while clean accuracy remains constant. When $\alpha$ exceeds 0.7, the high proportion of the original image's style embedding causes low-level visual features to be highly similar to the original, thereby reducing the model's ability to detect subtle changes and decreasing attack success. Conversely, when $\alpha < 0.5$, the model's capacity to detect subtle changes is enhanced due to an augmented fixed style contribution. However, this results in a substantial degradation of image quality and the creation of a significant disparity from the original. The experimental findings suggest that $\alpha = 0.5$ optimally balances effectiveness and stealthiness. The stylized image visualizations for various values of $\alpha$ are depicted in Fig. 10(c).

The mixing ratio $\beta$ regulates the fusion of original content and stylized trigger in poisoned images, directly impacting attack performance and stealthiness. Experimental results of various $\beta$ are shown in Fig. 11(a) and (b). The findings indicate that an augmentation in the style-image fusion ratio, signified by a higher $\beta$, results in a decline in attack success rate while maintaining clean accuracy at a consistent level. When $\beta > 0.7$, the high proportion of the original image obscures the stylized low-level visual features, causing fused poisoned images to resemble the original. This allows the model to associate natural features with target labels, reducing the success rate of the attack. Conversely, when $\beta < 0.5$, an increase in attack success is observed due to the presence of more stylized content. However, this also results in a degradation of image quality and the creation of a significant disparity from the original. A $\beta$ value of 0.7 is selected to optimize the balance between attack effectiveness and stealthiness. The visual outcomes of stylized images under varying $\beta$ are demonstrated in Fig. 11(c).

*3) The Impact of Different Loss:* In the training of the style transfer network, four losses were utilized: the backdoor loss, clean-label loss, style loss, and content loss. The total loss is expressed as $\mathcal{L}_{ssl} = \lambda_1\mathcal{L}_{bd} + \lambda_2\mathcal{L}_p + \lambda_3\mathcal{L}_s + \lambda_4\mathcal{L}_c$. Among these loss components, the backdoor loss $\mathcal{L}_{bd}$ is essential, as it serves as the fundamental driving force underlying the formation of backdoor attacks. To more accurately evaluate the effectiveness brought by different losses, we introduce the T-ACC metric, which is used to assess the classification accuracy of poisoned samples. It should be noted that under clean-label attacks, the higher the T-ACC, the better the performance. Next, on ImageNet-200, we conduct a series of experiments to assess the impact of other three loss components on the network's performance. The accuracy of clean image on benign model is 75.3% and the poisoning rate is 0.05%.

The results are presented in Table VII. When trained only with backdoor loss, ASR reaches its maximum value of 99.9%, but ACC and T-ACC are only 74.43% and 0.02, respectively. This indicates that the backdoor loss disrupts the relationship between poisoned samples and its categories, but does not establish a trigger and category association. This disruption is crucial for high efficiency under low poisoning rates. By suppressing the original features, the backdoor loss reduces the model's reliance on them, allowing the global style features to easily dominate the decision boundary with significantly fewer samples. When incorporating $\mathcal{L}_p$, there is a modest enhancement in the ACC and LPIPS, accompanied by a substantial decline in the ASR to 55.4%, which indicates that $\mathcal{L}_p$ helps enhance concealment, but can affect the availability of attacks. Although $\mathcal{L}_p$ encourages style images to be classified accurately, $\mathcal{L}_{bd}$ directly acts on poisoned images, forcing them to be classified incorrectly. Therefore, the effect of $\mathcal{L}_p$ on improving the classification accuracy of poisoned images is not significant, with T-ACC increasing by only 7 percentage points.

Next, $\mathcal{L}_{bd} + \mathcal{L}_p$ is adopt as the baseline loss function, and we then study the influence of consistency constraints, including style loss and content loss, on performance. Adding style loss $\mathcal{L}_s$ to the loss function results in to a slight decrease on the ACC and T-ACC, while adding $\mathcal{L}_c$ leads to enhancements across all four evaluation indicators. While both $\mathcal{L}_s$ and $\mathcal{L}_c$ enhance ASR from a baseline of 55.4%, $\mathcal{L}_s$ demonstrates a clearly superior effect, achieving an ASR of 99.7%, compared to 86.4% for $L_c$. On the other hand, $\mathcal{L}_c$ exhibits more favorable performance in terms of T-ACC and LPIPS, with T-ACC rising from 0.09 to 0.51 and LPIPS dropping from 0.178 to 0.142. These findings support our design motivation. Style loss weakens the association between clean samples and their true classes, thereby enhancing attack effectiveness, while the content loss establishes a stronger link between the trigger and the target class, improving both stealthiness and perceptual imperceptibility. Therefore, we incorporate both content loss and style loss to achieve a balanced trade-off between attack performance and evasion capability. Consequently, the joint use of both losses enables a more balanced attack strategy. As shown in the experimental results, combining both losses leads to the best performance in terms of ACC and T-ACC, while achieving suboptimal but still favorable results for ASR and LPIPS. The visualization results of various poisoned images are presented in Fig. 12 for intuitive comparison.


Fig. 12. The poisoned image of different loss.

TABLE VIII
THE INFLUENCE OF DIFFERENT MODEL ARCHITECTURES

| Dataset | Surrogate | Victim Model | | |
|---|---|---|---|---|
| | | ResNet34 | VGG16 | VGG19 |
| CIFAR-10 | ResNet18 | 92.63/98.87 | 94.02/96.54 | 94.21/97.63 |
| ImageNet-200 | ResNet18 | 75.51/94.31 | 77.19/98.92 | 77.64/96.36 |
| GTSRB | ResNet18 | 83.96/94.21 | 82.68/96.58 | 84.34/95.67 |

TABLE IX
THE INFLUENCE OF THE TRIGGER ON THE TRAINING LOSS

| Class | Clean | Method | | |
|---|---|---|---|---|
| | | DFB | Blend | Ours |
| Frog | 0.00407 | 0.8442 | 0.012 | **0.00402** |
| Cat | 0.00293 | 2.2070 | 0.031 | **0.00254** |
| Dog | 0.00275 | 2.1360 | 0.022 | **0.00372** |
| Truck | 0.00228 | 0.6456 | 0.026 | **0.00206** |
| Ship | 0.00236 | 0.7194 | 0.025 | **0.00390** |
| Horse | 0.00256 | 1.3612 | 0.041 | **0.00406** |

*4) Model Architecture and Target Class:* In order to verify the robustness of the proposed method to the network structure, modifications are made to the structures of the victim network and the decoder. This was done to verify the robustness of the attack method on different network architectures. The surrogate model is set to ResNet18, and the victim models are ResNet34, VGG16, and VGG19. As demonstrated in Table VIII, for all datasets, the victim models exhibit elevated ACC and ASR values. For instance, on the CIFAR-10 dataset, the ACC and ASR of ResNet34 achieve 92.63% and 98.87%, and on GTSRB, the ACC and ASR of VGG19 achieve 84.34% and 95.67%. Obviously, our trigger exhibits robust performance across various model architectures, thereby facilitating successful black-box attack scenarios.

To demonstrate that any category can be a target of backdoor attacks, we conducted targeted experiments on six different categories in the CIFAR-10 dataset and recorded the classification loss before and after poisoning for each category. Note that there are only 25 poisoned images each time. As presented in Table IX, owing to the low poisoning rate and the high quality of the poisoned data, the training loss of the backdoor model is comparable to that of the clean model and remains unaffected by the target class. In contrast, the training loss of the backdoor model trained using DFB [15] is significantly higher than that of the clean model. This is because DFB constructs triggers in a data free manner, which makes the triggers independent of the image content, resulting in significant differences between poisoned images and clean images. This further illustrates the stealthiness and stability of our attack, as the backdoor model's training process closely mimics that of a clean model regardless of the target class.

## V. Conclusion

This paper presents a novel approach to style-based clean-label backdoor attacks. The methodology utilizes style transfer to generate triggers, achieving decoupling of style and content. This ensures that poisoned images retain semantic content while associating specific styles with target labels. The method demonstrates its efficacy in balancing aggressiveness and stealthiness, thereby enhancing defense robustness. Experimental findings demonstrate that modifying 0.05% of the training data results in 99% attack success rate on target classes while maintaining unaltered clean accuracy. This approach surpasses advanced defenses such as NAD and CBD, as the triggers maintain the original features, thereby rendering feature-separation defenses ineffective. This work underscores the efficacy of style-based triggers in evading detection, thereby highlighting the necessity for more robust countermeasures against such attacks.

## VI. Ethical Considerations

The primary objective of this work is to expose a novel vulnerability in the machine learning supply chain—specifically, the stealthiness of style-based clean-label triggers—to facilitate the development of more robust defense mechanisms. By demonstrating that backdoors can be embedded via global style features without altering labels, we highlight a critical blind spot in current detection methods that focus on local pixel artifacts.

For model providers and defenders, this finding underscores the necessity of scrutinizing data sources not just for label noise but also for feature consistency. It calls for the development of advanced detection tools capable of identifying global statistical anomalies (e.g., Gram matrix distributions). We strictly condemn the use of this technique for malicious purposes. The code and datasets released with this paper are intended solely for academic research to assist the community in building safer and more trustworthy deep learning systems.

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